ECON 0100 | Exercise C1 | Externalities

Q1 | Smoked Toffee

Hogsmeade has seen a recent proliferation of candy shops selling smoked toffees. The proliferation of smoked toffee makers meant the town saw a subsequent rise in *smoke*. Wood smoke can cause lung issues. Public health officials have estimated the externality to be equal to carry a cost of 10 Coins to the health of Hogsmeade residents.

	a) A Single Exchange	**b) A Single Exchange**
	Buyer's MB: 60 Coins.	Buyer's MB: 80 Coins.
	Seller's MC: 55 Coins.	Seller's MC: 40 Coins.
Will buyer and seller exchange?	_____	_____
Total Surplus to buyer and seller:	_____	_____
Marginal Social Cost:	_____	_____
Total Surplus, including residents:	_____	_____
Deadweight Loss:	_____	_____

c) Market Equilibrium

Use a graph and algebra to find equilibrium price, quantity, and DWL. The demand curve and supply curve for smoked toffee can be represented using the following relationships.

$$D : P = 100 - Q_d \qquad S : P = 10 + Q_s$$

Equilibrium Price ($\hat{P}$): _____ Equilibrium Quantity ($\hat{Q}$): _____

Efficient Quantity (Q^*): _____ Deadweight Loss (DWL): _____

Q2 | Yard Flowers

As a small residential town, Hogsmeade has many shops and many homes near the town center. Residential homeowners often plant flowers in their front yards, contributing to the positive atmosphere in town. There are numerous garden stores selling supplies for planting and growing many varieties. The supply and demand relationships for a dozen flowers is given by the following equations, with prices in Coins.

$$D : P = 100 - \frac{1}{2}Q_d \qquad S : P = \frac{1}{2}Q_s$$

Recently, a study found a positive impact on the wellbeing of the community from the flowers planted in the front yards of their neighbors. The impact was estimated to carry a value of 10 Coins per dozen flowers planted. Use a graph and algebra to calculate the size of the market failure by identifying and calculating the deadweight loss.

Equilibrium Quantity ($\hat{Q}$): _____ Efficient Quantity (Q^*): _____

Deadweight Loss (DWL): _____